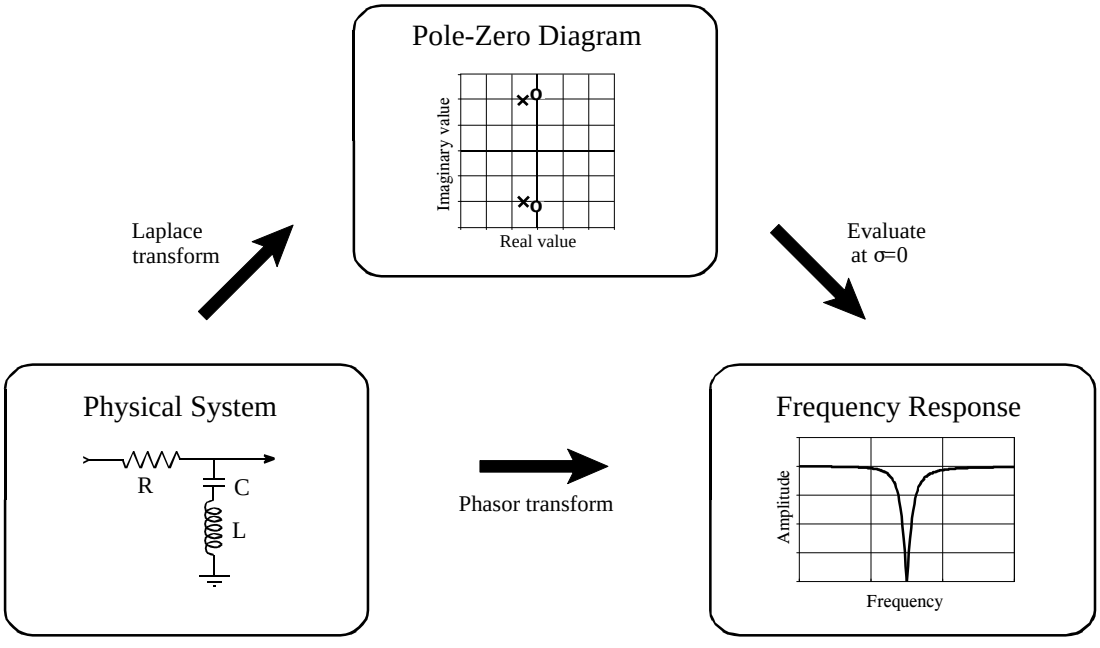


FIGURE 32-8
Strategy for using the Laplace transform. The phasor transform presented in Chapter 30 (the method using R, $j\omega L$, & $-j/\omega C$) allows the frequency response to be directly calculated from the parameters of the physical system. In comparison, the Laplace transform calculates an s-domain representation from the physical system, usually displayed in the form of a pole-zero diagram. In turn, the frequency response can be obtained from the s-domain by evaluating the transfer function along the imaginary axis. While both methods provide the same end result, the intermediate step of the s-domain provides insight into why the frequency response behaves as it does.

cancel, providing a unity frequency response at low frequencies. The situation doesn't change significantly until the ant moves near the pole and zero location. When approaching the zero, the value of $H(s)$ drops suddenly, becoming *zero* when the ant is upon the *zero*. As the ant moves past the pole and zero pair, the value of $H(s)$ again returns to unity. Using this type of visualization, it can be seen that the width of the notch depends on the distance between the pole and zero.

Figure 32-8 summarizes how the Laplace transform is used. We start with a physical system, such as an electric circuit. If we desire, the phasor transform can directly provide the frequency response of the system, as described in Chapter 30. An alternative is to take the Laplace transform using the four step method previously outlined. This results in a mathematical expression for the transfer function, $H(s)$, which can be represented in a pole-zero diagram. The frequency response can then be found by evaluating the transfer function along the imaginary axis, that is, by replacing each s with $j\omega$. While both methods provide the same result, the intermediate pole-zero diagram provides an understanding of *why* the system behaves as it does, and how it can be changed.